SYNOPSIS

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Project Title: Deep Feature Optimization Framework for Early Breast Cancer Detection

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Abstract

Breast cancer is one of the leading causes of cancer-related deaths among women worldwide, and early diagnosis is essential for effective treatment and improved survival rates. This work proposes an integrated framework combining transfer learning-based deep feature extraction, meta-heuristic feature selection, and traditional machine learning classifiers for efficient breast cancer histopathological image classification. Deep features are extracted using the pre-trained ResNet18 model, generating 512 features per image. Three optimization algorithms — PSO, ASO, and EO — are applied to select the most relevant feature subset, which is then classified using SVM, KNN, and Decision Tree classifiers. Experimental results on the BreakHis dataset confirm that the EO-SVM combination at 200X magnification achieves the best performance of 97.73% accuracy, demonstrating an efficient and accurate solution for automated breast cancer diagnosis.

Specific Contribution

I contributed to Module 1 — Data Preparation and Deep Feature Extraction. My responsibilities included collecting the BreakHis dataset, performing image preprocessing such as resizing and normalization, implementing the ResNet18 model using transfer learning, extracting 512 deep feature vectors from histopathological images, and generating the feature dataset required for subsequent optimization and classification stages.

Specific Learning

Through this work, I learned how to handle and preprocess medical imaging datasets, including tasks such as image resizing, normalization, and structuring class-wise data for deep learning pipelines. I gained a thorough understanding of transfer learning using pre-trained CNN architectures, specifically ResNet18, and how to leverage pre-trained weights to extract meaningful deep features from limited medical data. I understood the importance of global average pooling in generating compact feature representations from convolutional layers. I also learned how high-dimensional feature vectors (512 features) impact downstream classification tasks and why dimensionality reduction is necessary. This module gave me practical exposure to building end-to-end feature extraction pipelines using Python, TensorFlow, Keras, and NumPy, and deepened my appreciation for the role of data quality and preprocessing in determining model performance.

Technical Limitations & Ethical Challenges faced

During the data preparation phase, a key technical challenge was ensuring consistent image quality across all four magnification levels (40X, 100X, 200X, 400X) of the BreakHis dataset. Variations in staining intensity, image resolution, and background artifacts introduced noise that affected the reliability of extracted features. Loading and preprocessing thousands of high-resolution histopathological images was also computationally intensive and required efficient memory management. From an ethical perspective, working with medical imaging data requires careful handling to ensure patient privacy is maintained, even though the BreakHis dataset is publicly available and anonymized. There is also a concern about fairness in model performance — if the training data does not adequately represent all patient demographics, the resulting model may not generalize well to diverse clinical populations. Additionally, deploying automated systems for cancer diagnosis raises questions about clinical accountability and the importance of keeping medical professionals in the decision-making loop.

Keywords: Dataset Preparation, Image Preprocessing, Transfer Learning, ResNet18, Deep Feature Extraction, BreakHis Dataset, Histopathological Image Analysis

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Date: